NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Computer Science & Information Technology
BS- ARTFICIAL INTELLIGNECE

PROJECT REPORT

FaultSense AI:

Hybrid Symbolic & XGBoost Telemetry Engine for Industrial Equipment Diagnostics

A Hybrid Artificial Intelligence Framework for Real-Time Industrial Equipment Diagnostics and Role-Based Action Protocols

PROJECT DETAILS

Subject	KNOWLEDGE BASE SYSTEM
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2. Abstract

Modern industrial and automated manufacturing environments rely heavily on complex CNC machinery and automated assembly lines. Unexpected equipment failures can cause catastrophic financial losses, prolonged operational downtimes, and severe safety hazards. FaultSense AI addresses these critical challenges by providing an intelligent web-based diagnostic portal. The system ingests operational sensor data, processes it through a hybrid architecture combining machine learning (XGBoost) with a domain-specific symbolic rule engine, and delivers immediate, role-customized diagnostic reports. This report details the comprehensive architecture, dataset features, telemetry parameters, inference mechanics, system workflows, and operational impact of FaultSense AI.

3. Introduction

3.1 Background

In high-precision manufacturing plants, heavy machinery such as milling stations operates under extreme thermal, mechanical, and electrical stresses. Over time, components experience structural wear, fluid loss, electrical overloads, or thermal saturation. Traditional monitoring techniques often depend on reactive maintenance—fixing equipment only after a breakdown occurs—or manual threshold checks that are prone to human oversight.

3.2 Overview of FaultSense AI

FaultSense AI bridges this gap by acting as a proactive predictive maintenance dashboard. It evaluates live machine sensor readings, detects multiple failure modes simultaneously, determines safety criticality levels, and prescribes targeted corrective actions tailored to specific plant stakeholders (Floor Operators, Maintenance Technicians, and Plant Managers).

4. Problem Statement & Proposed Solution

4.1 Problem Highlights

In modern manufacturing plants, manual monitoring of complex CNC machinery telemetry often leads to delayed fault detection, unexpected equipment downtime, and severe safety hazards. Traditional systems fail to process high-dimensional sensor data efficiently or provide actionable, role-specific maintenance protocols, necessitating an automated, intelligent diagnostic framework.

- **High Data Complexity:** Raw industrial telemetry data (such as temperatures, rotational speeds, and torque values) is complex and difficult for human operators to continuously interpret in real time.
- **Delayed Fault Identification:** Sub-surface failures—such as Heat Dissipation Failures (HDF) or Power Failures (PWF)—often escalate before manual inspection catches them, leading to major structural damage and unexpected downtime.
- **Lack of Role-Specific Protocols:** Generic system error codes fail to provide clear, actionable steps tailored to different hierarchical personnel levels within a manufacturing plant.

4.2 Proposed Solution

To address these operational bottlenecks, FaultSense AI provides an intelligent, web-based diagnostic portal that automates industrial equipment health monitoring. The system bridges the gap between raw telemetry and maintenance execution by seamlessly ingesting operational inputs, processing them through a hybrid decision framework (integrating machine learning capabilities like XGBoost with a deterministic knowledge base and inference engine), and instantly outputting comprehensive diagnostic reports. Instead of manual guesswork, plant personnel receive

> **clear status classifications, risk assessments, and targeted action workflows tailored specifically to their operational roles, ensuring rapid response and asset protection.**

4.3 Key Features

Table 4.1: Key Features of FaultSense AI

No.	Feature	Description
01	**Interactive Telemetry Controls:**	*Responsive UI components featuring real-time sliders and machine quality variant selections.*
02	**Hybrid Symbolic + XGBoost Inference**	*Combines probabilistic machine learning models with deterministic rules to achieve high diagnostic confidence and accuracy.*
03	**Knowledge-Base-Driven Explanations**	*Employs an expert rule system to trace exact sensor triggers, ISO standards, and technical definitions.*
04	**Role-Based Action Protocols**	*Customized maintenance workflows filtering down to Floor Operators, Maintenance Technicians, and Plant Managers.*
05	**Interactive Diagnostic Dashboard**	*Capability to identify and report concurrent anomalies (e.g., simultaneous Power Failures and Overstrain Failures) using model, inference engine, and knowledge base.*
06	**Downloadable JSON Diagnostic Reports**	*Instant export functionality for saving structured log files for auditing and record-keeping.*

5. Dataset & Parameter Explanations

5.1 Dataset Overview

UDI	Product ID	Type	Air temperature [K]	Process temperature [K]	Rotational speed [rpm]	Torque [Nm]	Tool wear [min]	Machine failure	TWF	HDF	PWF	OSF	RNF
1	M14860	M	298.1	308.6	1551	42.8	0	0	0	0	0	0	0
2	L47181	L	298.2	308.7	1408	46.3	3	0	0	0	0	0	0
3	L47182	L	298.1	308.5	1498	49.4	5	0	0	0	0	0	0
4	L47183	L	298.2	308.6	1433	39.5	7	0	0	0	0	0	0
5	L47184	L	298.2	308.7	1408	40	9	0	0	0	0	0	0
6	M14865	M	298.1	308.6	1425	41.9	11	0	0	0	0	0	0
7	L47186	L	298.1	308.6	1558	42.4	14	0	0	0	0	0	0
8	L47187	L	298.1	308.6	1527	40.2	16	0	0	0	0	0	0
9	M14868	M	298.3	308.7	1667	28.6	18	0	0	0	0	0	0
10	M14869	M	298.5	309	1741	28	21	0	0	0	0	0	0
11	H29424	H	298.4	308.9	1782	23.9	24	0	0	0	0	0	0
12	H29425	H	298.6	309.1	1423	44.3	29	0	0	0	0	0	0
13	M14872	M	298.6	309.1	1339	51.1	34	0	0	0	0	0	0
14	M14873	M	298.6	309.2	1742	30	37	0	0	0	0	0	0
15	L47194	L	298.6	309.2	2035	19.6	40	0	0	0	0	0	0
16	L47195	L	298.6	309.2	1542	48.4	42	0	0	0	0	0	0
17	M14876	M	298.6	309.2	1311	46.6	44	0	0	0	0	0	0
18	M14877	M	298.7	309.2	1410	45.6	47	0	0	0	0	0	0
19	H29432	H	298.8	309.2	1306	54.5	50	0	0	0	0	0	0
20	M14879	M	298.9	309.3	1632	32.5	55	0	0	0	0	0	0
21	H29434	H	298.9	309.3	1375	42.7	58	0	0	0	0	0	0
22	L47201	L	298.8	309.3	1450	44.8	63	0	0	0	0	0	0
23	M14882	M	298.9	309.3	1581	30.7	65	0	0	0	0	0	0
24	L47203	L	299	309.4	1758	25.7	68	0	0	0	0	0	0
25	M14884	M	299	309.4	1561	37.3	70	0	0	0	0	0	0
26	L47205	L	299	309.5	1861	23.3	73	0	0	0	0	0	0

Table 5.1: Dataset Summary

Dataset Name	*AI4I 2020 Predictive Maintenance Dataset*
Source	*Kaggle*
Number of Records	10150
Number of Features	14
Target Variables	*Binary Machine Failure Flag (0/1) & Specific Failure Modes (TWF, HDF, PWF, OSF, RNF)*
Class Distribution	*Imbalanced dataset representing real-world industrial normal operations vs. rare failure events*
Pre-processing Steps	*Cleaning, encoding, scaling, balancing, model training*

5.2 Core Sensor Parameters

Table 5.2: Core Sensor Parameters

Parameter	Unit	Description	Operating Range
Air Temperature	*[K]*	*Ambient room temperature baseline measured via an infrared probe at the CNC intake exhaust vent.*	273-350 K
Process Temperature	*[K]*	*Working operational heat logged via a digital thermal laser at the active tool-spindle contact point.*	273-350 K
Rotational Speed	*[rpm]*	*Spindle rotational speed evaluated from the active drive readout tachometer under load conditions.*	0-4000rpm
Torque	*[Nm]*	*Mechanical rotational force and twisting load measured via digital current load transducers.*	0-100Nm
Tool Wear	*[min]*	*Cumulative cutting duration elapsed for the active tool bit tracked against preventive thresholds.*	0-300min
Product Type	*[L / M / H]*	*Machine quality variant tiers (Lite, Medium, High) defining unique structural tolerance limits*	*[—]*

5.3 Identified Fault Types & Error Modes (HDF, PWF, OSF, TWF, RNF)

Table 5.3: Fault Modes Detected by FaultSense AI

Code	Fault Mode	Description	Detection Logic
HDF	Heat Dissipation Failure	*Thermal saturation caused by fluid loss or convective cooling blockages in the liquid loop.*	*Temperature difference (Process Temp - Air Temp) < 2.0K AND Spindle speed < 1380 RPM*
PWF	Power Failure	*Electrical drive anomalies or torque stalls where mechanical power falls outside safe physical bounds.*	*Mechanical Power (Torque multiplied by Angular Velocity) < 3500 Watts OR > 9000 Watts*
OSF	Overstrain Failure	*Structural stress resulting from excessive mechanical feed loads combined with degraded tool geometry.*	*Strain Product (Tool Wear multiplied by Torque) > Variant Limit (11000 for Lite, 12000 for Medium, 13000 for High quality)*
TWF	Tool Wear Failure	*Abrasive friction accumulation exceeding variant-specific cutting limits violating ISO 15641.*	*Tool wear minutes > Variant Limit (Lite: > 240 mins, Medium: > 210 mins, High: > 220 mins)*

Code	Fault Mode	Description	Detection Logic
RNF	Random Failure	*Stochastic, unpredicted component breakdowns occurring due to rare operational anomalies.*	*ML Model probability classification trigger / Baseline anomaly*

Data Pre-processing & Feature Engineering Summary: -

Data pre-processing was systematically performed to ensure high data integrity and model reliability. Initial steps involved checking for missing values, handling duplicates, and filtering outliers within valid sensor physical bounds. To prepare the categorical machine variant attributes (L, M, H) for machine learning compatibility, One-Hot Encoding was applied. Furthermore, domain-specific feature engineering was executed to enrich the feature space, which included calculating the thermal differential ({Process Temp} - {Air Temp}), mechanical power output (Torque * Angular Velocity), and cumulative strain products (Tool Wear * Torque), enabling the model and knowledge base to accurately capture complex industrial failure patterns.

Steps of data preprocessing:-

1.Load and Inspect

Load the messy AI4I 2020 data. Check shape, types, nulls and duplicates. Repeated UDI rows and missing machine types show up.

2. Explore

Plot boxplots, histograms, a correlation heatmap and failure counts. Failures are rare, so the classes are heavily imbalanced.

3. Clean and Audit

Drop duplicates and merge repeated UDIs. Recover missing types from the Product ID. Compare Machine failure with the five failure-mode flags. Save mismatched rows for the rule engine.

4. Split and Fix Values

Do an 80/20 stratified split. Turn impossible sensor readings into missing values. Fill all gaps with the training median, so nothing leaks from the test set.

5. Features

Add temperature difference, power, overstrain index and torque-speed ratio. One-hot encode machine type. Scale inputs for Logistic Regression.

6. Train and Compare

Train four models and evaluate each on the test set (table below).

Model	Precision	Recall	F1	ROC-AUC	PR-AUC	FP	FN
Logistic Regression	0.17	0.85	0.28	0.926	0.362	287	10
Random Forest	0.62	0.74	0.67	0.972	0.773	31	18
XGBoost (baseline)	0.83	0.78	0.80	0.949	0.830	11	15
LightGBM	0.80	0.76	0.78	0.947	0.832	13	16
XGBoost (tuned)	**0.86**	**0.79**	**0.82**	**0.965**	**0.839**	**9**	**14**

7. Tune and Deploy

Randomized search with 5-fold CV tunes XGBoost for F1. The tuned model has the best precision, recall, F1 and PR-AUC, and the fewest false alarms and misses. It is served by the FastAPI backend.

Structure and design of knowledge base

- **Dictionary-Based Architecture:** The knowledge base is structured as a centralized Python dictionary (KNOWLEDGE_BASE) where each industrial failure mode (such as TWF, HDF, PWF, OSF, and RNF) serves as a primary key.
- **Core Engineering Metadata:** Each entry maps directly to standardized industrial metadata, including a unique Failure ID (Failure_ID), human-readable failure name (Failure_Mode_Name), affected hardware component (Target_Component), and official compliance standard references (e.g., ISO 15641, ISO 13374, IEC 60204-1).
- **Technical Engineering Core:** It explicitly defines physical behaviors, containing field definitions, observed operational symptoms, and precise mathematical trigger conditions required to flag an anomaly.
- **Variant Profiles (L, M, H):** The knowledge base encodes multi-tier quality profiles for Lite (L), Medium (M), and High (H) machine grades, establishing dynamic parameters such as maximum tool wear limits, thermal sensitivities, and structural yield thresholds.
- **Multi-Stakeholder Action Matrix:** To satisfy operational safety protocols, it houses a role-based action matrix that defines distinct step-by-step responses for Floor Operators, Maintenance Technicians, and Plant Managers.
- **System Constraints Matrix:** It stores impact assessment metrics for every fault mode, detailing safety criticality levels (Level-1 to Level-5), expected downtime windows, financial cost indices, and production continuity risks.

Structure and Logic of the Inference Engine

- **Model Integration & Path Configuration:** The engine dynamically loads the pre-trained tuned XGBoost classifier (best_xgb_tuned.joblib) and feature mapping arrays to ensure seamless compatibility with incoming telemetry inputs.
- **Input Validation & Missing Data Handling** : It verifies all incoming sensor parameters (Air Temperature, Process Temperature, Rotational Speed, Torque, Tool Wear) against strict physical safety ranges (VALID_RANGES). It identifies missing fields or out-of-bound anomalies and extracts machine variant types (L, M, H) directly from product identifiers if necessary.
- **Derived Feature Engineering:** Before passing data to the machine learning model, the script calculates essential industrial metrics:
    - Temp_diff, Power , Overstrain Index , Torque-Speed Ratio
- **Independent Forward-Chaining Rule Verification:** Running parallel to the machine learning model, the engine executes strict, non-blocking conditional statements to evaluate rule-based triggers independently:
    - *Tool Wear Failure (TWF):* Triggers when tool wear exceeds variant-specific limits (Max_Wear_Limit).
    - *Heat Dissipation Failure (HDF):* Triggers when thermal difference is below 2.0K and rotational speed is below 1380 RPM.
    - *Power Failure (PWF):* Triggers when mechanical power drops below 3500W or exceeds 9000W.
    - *Overstrain Failure (OSF):* Triggers when the strain product exceeds variant-specific structural yield limits .
- **State Fusion & Fallback Logic:** The engine aggregates outputs from both the ML model and symbolic rules . If rules detect active anomalies, the status is forced to "Fault Detected". If the ML model flags an anomaly without matching a deterministic rule, it defaults to a random anomaly code (RNF).
- **Knowledge Mapping:** For every detected fault code, it queries the Knowledge Base (KB) to fetch respective ISO references, technical definitions, action protocols, and constraint matrices into a unified dictionary map.

Mechanics of the Explanation Generator:-

- **Multi-Dimensional Text Synthesis**: **The function (generate_comprehensive_explanation) translates raw inference dictionaries and sensor facts into structured, human-readable assessments.**
- **Condition-Based Branching: If the system status is normal, it outputs a baseline confirmation. If faults are active, it dynamically constructs targeted narratives across five core analytical pillars:**
    1. **What: Identifies active fault codes and affected target components.**
    2. **Why: Formulates contextual rule explanations citing exact sensor readings and variant parameters (e.g., thermal thresholds, power wattage, or wear durations).**
    3. **How Sure: Reports hybrid diagnostic confidence percentages combining ML probabilities and rule verification.**
    4. **Impact: Summarizes operational downtime, financial risk index, and safety levels.**
    5. **Action Protocols: Formats role-based directives into a clean string separating operator, technician, and manager responsibilities**

WORLFLOW OF EXPERT SYSTEM:-

- **Telemetry Input:** The user provides operational sensor readings (temperatures, speed, torque, tool wear) and machine variant types through the frontend interface.
- **Feature Engineering & ML Prediction:** The inference engine calculates derived metrics (like thermal differences and power), and passes it to the trained XGBoost model to predict whether the machine is operating normally or heading toward a failure, along with a confidence score.
- **Symbolic Rule Verification (Knowledge Base):** the system checks physical rules and mathematical thresholds defined in the Knowledge Base (such as specific limits for HDF, PWF, OSF, and TWF) to ensure physical laws are satisfied.
- **Result Fusion & Data Lookup:** The engine fuses the ML prediction with the rule-based flags. If an anomaly is verified, it fetches the corresponding ISO standards, component details, risk metrics, and role-specific action protocols from the Knowledge Base.
- **Explanation Generation & Reporting:** Finally, the Explanation Generator transforms all structured findings into a clear, multi-dimensional assessment detailing What was found, Why it happened, How sure the system is, the operational Impact, and step-by-step Action Protocols for operators, technicians, and managers.

6 System Architecture & Working Methodology

6.1 End-to-End Workflow & Flow Diagram

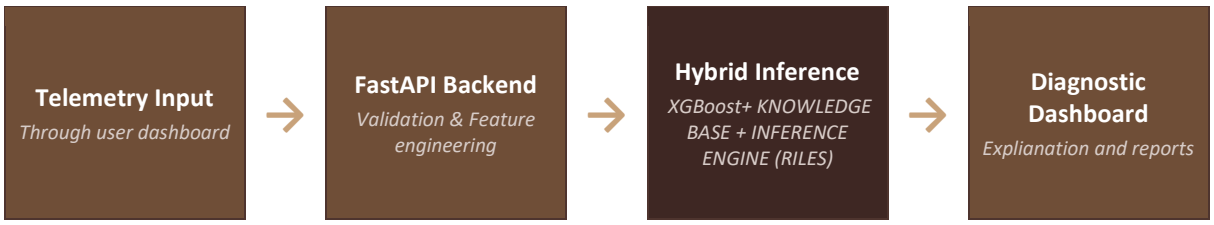


Figure 6.1:

OUTPUT EXPLIANATION:-

1. Status: Shows whether a machine is operating normally or a fault has been detected.

2. Confidence Probability: Displays the system's confidence in its diagnostic prediction as a percentage.

3. Detected Fault Codes: Lists the fault codes identified by the system, such as HDF, PWF, and OSF. Multiple faults can be detected together.

4. Explanation: Provides a brief summary of the anomalies verified by the hybrid system.

5. Failure Mode: Identifies the type of failure detected, such as Heat Dissipation Failure, Power Failure, or Overstrain Failure.

6. Target Component: Identifies the specific machine component associated with the detected fault.

7. ISO Standard: Specifies the industrial standard associated with the detected fault.

8. Technical Definition: Explains the technical cause and nature of the identified failure.

9. Action Protocols: Provides recommended actions for three responsible roles:
- Floor Operator Action: Immediate steps to ensure machine and workplace safety.
- Maintenance Technician Action: Technical inspection and troubleshooting procedures.
- Plant Manager Action: Maintenance decisions and production management actions.
10. Constraints Matrix: Describes the operational consequences and limitations associated with the fault:
- Safety Criticality: Indicates the level of risk to personnel and equipment.
- Downtime Impact: Estimates the time required for maintenance or recovery.
- Financial Cost Index: Indicates the expected maintenance or repair cost.

- Production Continuity: Explains how the fault affects ongoing production.
11. Summary Card: Combines the detected fault, affected component, triggering condition, confidence, operational impact, and immediate action into one diagnostic summary.

12. What: Describes which fault was detected and which component is affected.

13. Why: Explains the sensor conditions or rule-based thresholds that triggered the diagnosis.

14. How Sure: Shows the system's confidence in the diagnostic result and mentions its verification through the ML model and rule engine.

15. Impact: Summarizes the fault's safety risk, expected downtime, and financial consequences.

16. Action: Combines the recommended actions for the operator, technician, and manager.

9. Limitations

Despite its capabilities, FaultSense AI has the following limitations:

1. Single Synthetic Dataset: The model is trained and evaluated only on the AI4I 2020 dataset, which is simulated and may not fully represent the noise and variability of real industrial machines.

2. Limited Fault Classes: The system covers five failure modes: TWF, HDF, PWF, OSF, and RNF. RNF is a fallback label rather than a physically defined fault; therefore, other real-world failure types are not currently diagnosed.

3. No Live Sensor Integration: Sensor readings are entered manually through the web interface. The system is not connected to PLC, SCADA, or IoT sensors and does not support real-time machine monitoring.

4. Class Imbalance and Missed Failures: Failure cases represent approximately 3% of the dataset. The tuned model achieves approximately 79% recall, meaning some actual failures may remain undetected.

5. Fixed Rule Thresholds: Knowledge-base thresholds, such as tool-wear and power limits, are manually configured and have not been calibrated using real industrial plant data.

6. No Data Persistence or User Management: Diagnoses are not stored in a database, and the system does not currently provide login functionality or maintenance history. Diagnostic records are retained through downloadable JSON reports.

11. Future Enhancements

The following enhancements are proposed to further improve the functionality, reliability, and practical applicability of FaultSense AI.

1. Live PLC / IoT Sensor Integration: Integrate directly with machine sensors to receive readings for real-time monitoring and automatic diagnosis instead of manual input.

2. Validation on Real Industrial Data: Retrain and evaluate the model using real industrial data and calibrate knowledge-base thresholds to improve reliability.

3. SHAP-Based Explanations: Introduce feature-level explanations to show how individual sensor readings contribute to model predictions.

4. Maintenance History Database and User Login: Store previous diagnoses and maintenance actions while providing role-based access for operators, technicians, and managers.

5. Remaining Useful Life Prediction: Estimate the remaining operating life of tools or components to support predictive maintenance planning.